Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

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Abstract

Between 70–96 % of maritime casualties involve human error [1], and heavy weather makes coordination harder by degrading both perception and bridge-to-bridge radio. We present a tick-based agent-based model (ABM) of Baltic and North Sea traffic. Vessels follow depth- and lane-constrained routes built from open bathymetry and Automatic Identification System (AIS) density data. Encounters are resolved with a Closest Point of Approach (CPA) check and an International Regulations for Preventing Collisions at Sea (COLREGs)-style give-way rule. The main modelling choice is that every collision warning succeeds only with probability equal to the product of a local weather factor and a crew-fatigue factor. Collision damage uses Technical University of Denmark (DTU) SIMCOL [2]; route-network collision frequency is checked with the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) Waterway Risk Assessment Program (IWRAP) Mk II [3]; rescue follows International Aeronautical and Maritime Search and Rescue (IAMSAR) practice [4] with the MASSIM maritime-search simulation and Koopman random search [5, 6] and Ashrafi seasonal degradation [7]. We compare a classical baseline, a weather-aware baseline, and a fatigue-aware Proposed System across four scenarios (30 seeds each; 360 runs) with Mann–Whitney U tests, Holm–Bonferroni correction, and Cliff's δ. Pre-registered fatal-rate and survival thresholds (H1–H3) were not confirmed after correction. Preparedness P_{prep} rose with large confirmed effects in every scenario ($|\delta| \geq 0.90$). Mean Storm Corridor collisions fell by about 28 % versus Baseline A (δ=0.32, not significant after Holm).

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport moves most of world trade, but safety remains limited by human factors [1]. The European Maritime Safety Agency (EMSA) [8] reports weather as a frequent compounding factor in serious casualties, especially in congested coastal waters, and hull and machinery claims remain large [9]. The International Regulations for Preventing Collisions at Sea (COLREGs) [10] assume that crews see the encounter and can agree who gives way. That assumption breaks down under two effects that many maritime agent-based models (ABMs) treat only loosely [11, 12, 13]. Weather and radio: sea clutter and storms reduce the chance that a marine very high frequency (VHF) warning is heard and acted on, yet many models assume that once risk is detected the manoeuvre always happens. Crew fatigue: fatigue builds over a watch and follows a circadian pattern [14], but prior work often keeps it in a side model without a direct link to whether a warning is executed. This paper couples those effects in one product: warning success depends on both local weather and crew fatigue (Fig. 1). Sections 3–6 define the Baltic–North Sea ABM, fatigue-aware Closest Point of Approach (CPA) avoidance, DTU SIMCOL damage [2], and search and rescue (SAR). Section 7 places network frequency on an IWRAP scale [3]; Sections 8–10 report the factorial comparison.

2. Objectives and Hypotheses

We test whether linking fatigue management to warning reliability reduces fatal outcomes more than classical, weather-unaware practice. We implement three configurations (storm-zone radio loss is shared by all methods—environmental, not a method privilege). Baseline A (Classical) uses COLREGs give-way with no weather-aware slowdown and unmanaged fatigue. Baseline B (Shore Rest Alerts) adds weather-aware speed reduction and slower fatigue build-up. The Proposed System keeps Baseline B's slowdowns and adds smart watch scheduling (lowest fatigue build-rate). Hypotheses: H1—Proposed reduces the fatal-event rate (Table 3, key performance indicator (KPI) 1) by $\geq$20 % versus Baseline A ($p < 0.05$, Mann–Whitney U [15], Cliff's $|\delta| \geq 0.2$ [16]); H2—Proposed improves post-incident survival (KPI 4) by $\geq$15 % versus Baseline A; H3—under Storm Corridor, Proposed beats Baseline B on the fatal-event rate ($p < 0.05$).

3. Simulation Model

3.1. Domain and Time

The domain is the Baltic and North Sea: latitude $\phi \in [50.5°, 66.0°]$ N and longitude $\lambda \in [-5.0°, 31.0°]$ E. An equirectangular projection map with a mid-latitude correction sets one field unit to one nautical mile at $\phi_0 = 58.25°$:

$$x = (\lambda - \lambda_{\min})\, k_{\mathrm{lon}}, \qquad y = (\phi - \phi_{\min})\, k_{\mathrm{lat}},$$

with $k_{\mathrm{lat}} = 60\,\mathrm{nm}/°$ and $k_{\mathrm{lon}} = 60 \cos\phi_0$. East–west distortion at the box edges is about 6–10 % and is left uncorrected; distances inside the engine are Euclidean nautical miles. Coastline polygons from Natural Earth sit in an R-tree spatial index[1] for land and grounding tests. Bathymetry is used only offline for route planning (Section 3.3). Time steps are $\Delta t = 15$ min (0.25 h); the default run length is $T = 2880$ ticks (30 days). A wall-clock variable starts at 06:00 Coordinated Universal Time (UTC) and drives the circadian fatigue term (Section 4). Each

[1] Overview: https://en.wikipedia.org/wiki/R-tree

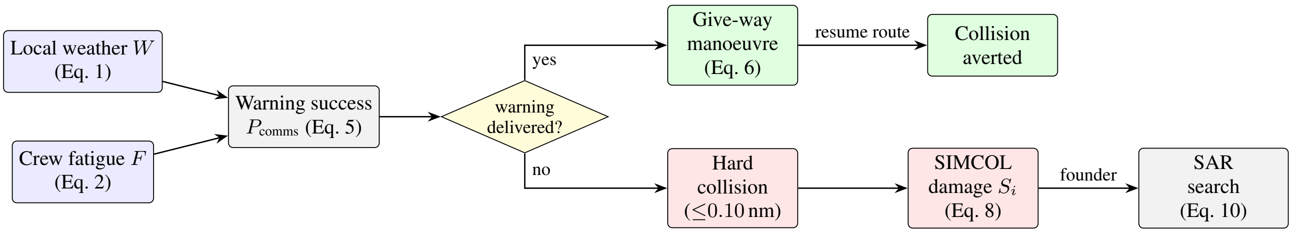

Figure 1: *Causal pipeline. Weather W and fatigue F set P_{comms}. A delivered warning yields a give-way manoeuvre; a missed warning leaves the encounter to geometry. Hard collisions feed SIMCOL; foundering enters SAR.*

run uses one seeded `SmallRng`, so the same seed yields identical KPI logs.

3.2. Vessels and Voyage Logic

Agents are ships; after an incident, rescue assets appear (Section 6). Ship i has position $\mathbf{p}_i$, waypoint route R_i, travel direction $d_i \in \{+1, -1\}$, heading ψ_i, cruise speed v_i, state $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$, crew $n_i = 10$, and fatigue $F_i \in [0, 1]$. Routes come from the offline network of Section 3.3 (`data/ais_paths.json`); ports are polygons in `data/ports.json`. Entering a port docks the ship. Cruise speed is drawn at spawn ($u \sim \mathcal{U}(0, 1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / roll-on/roll-off (ro-ro, ro-pax)} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases}$$

Waypoint pursuit: the active target is an avoidance waypoint if one is queued, otherwise the current route waypoint. A waypoint is reached within 0.5 nm. The step length is $s = \min(v_i\,\Delta t,\ \rho)$. Moves use up to five geometrically halved lengths so thin land is not tunnelled; a post-move check reverts grounding to the last valid position. At a port or route end the ship dwells $D \sim \mathcal{U}\{8, 48\}$ ticks, then reverses. Fleet size N is held fixed: when a ship is removed, a new one is spawned on a random synthesised track. Same-destination following: co-directional ships with endpoints within 5 nm share a destination; a trailer within 3 nm of a leader matches the leader's speed, and within 0.5 nm it holds station. This avoids side-by-side overlap that the give-way rule does not separate.

3.3. Route Generation

Traffic geometry is produced offline so experiments stay reproducible. Depth comes from the European Marine Observation and Data Network (EMODnet) digital terrain model (DTM) [17]. Lane attractiveness comes from EMODnet vessel-density maps [18, 12]. Both are fetched once via the Open Geospatial Consortium (OGC) Web Coverage Service (WCS) and stored as grids. Ports are a gazetteer of 42 harbours snapped to navigable water. Each class c has draft T_c and under-keel clearance (UKC_c) [19]. A cell of depth h is navigable iff $h \geq T_c + \text{UKC}_c + m_s$ with buffer $m_s = 2$ m (Table 1). Deep-draft very large crude carriers (VLCCs) are excluded (Danish Straits).

Between sampled port pairs, a weighted A* pathfinding planner [20] runs on a 1 nm grid. Cell cost uses

$$\mu = 1 + w_h\, s_h + w_d\, s_d + w_\ell\,(1 - a_\ell),$$

Table 1: *Minimum transit depth by vessel class (draft + under-keel clearance).*

Class	Draft T_c (m)	UKC_c (m)	Required h (m)
Passenger	6.5	1.5	8
Cargo	11.0	2.0	13
Tanker	14.0	3.0	17

penalising shallow margins (s_h), proximity to shore/shoal (s_d), and off-lane cells (a_ℓ). Default weights are $(w_h, w_d, w_\ell) = (3, 4, 2.5)$. Paths are simplified under a depth constraint and checked against raw bathymetry every 0.15 nm. Seeded cost jitter keeps outputs deterministic per seed. The default network has 80 routes (35 cargo, 25 passenger, 20 tanker).

3.4. Weather Field

A 50×50 grid holds sea state s, visibility v, and wind u in $[0, 1]$. They fuse into hazard $W_{ij} \in [0, 1]$:

$$W = \text{clip}\left(\frac{w_s\, s + w_v\,(1 - v) + w_u\, u}{w_s + w_v + w_u},\ 0,\ 1\right), \tag{1}$$

with $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. Each ship samples the nearest cell. The field advances each tick through a calm/storm regime chain, Lagrangian storm cells, first-order autoregressive (AR(1)) background fields, overlays, coupling, and smoothing (parameters in Appendix A). Independently, Storm Corridor places a moving storm of radius r_{storm} that drifts at 0.30 nm/tick along a fixed track (English Channel $\rightarrow$ North Sea $\rightarrow$ Skagerrak $\rightarrow$ Kattegat $\rightarrow$ Baltic $\rightarrow$ Gdańsk). Inside the zone, link reliability and non-A speeds are degraded (Sections 4 and 9.2; Table 2).

4. Crew Fatigue and Collision Avoidance

4.1. Fatigue Dynamics

Fatigue $F_i \in [0, 1]$ updates every tick. The circadian drive peaks near 03:00:

$$c(t_{\text{utc}}) = \tfrac{1}{2}\Big(1 - \cos\big(\tfrac{2\pi(t_{\text{utc}} - 3)}{24}\big)\Big) \in [0, 1].$$

While ACTIVE, with local hazard W,

$$\Delta F = \big(\beta_{\text{base}} + \beta_W\, W + \beta_c\, c(t_{\text{utc}})\big)\, m(\text{method}), \tag{2}$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015,\ 0.0040,\ 0.0008)$ and

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A} \\ 0.82 & \text{Baseline B} \\ 0.65 & \text{Proposed,} \end{cases} \tag{3}$$

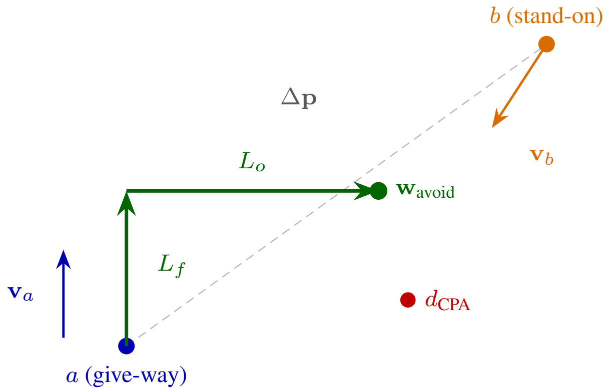

Figure 2: *Give-way geometry. Ship a steers to a temporary waypoint $\mathbf{w}_{avoid}$ placed L_f ahead and L_o to starboard (Eq. 6).*

then $F_i \leftarrow \min(1, F_i + \Delta F)$. While DOCKED, recovery is 0.025 per tick. Above threshold $F^* = 0.40$, fatigue reduces warning reliability with maximum penalty $\kappa = 0.65$:

$$g(F_i) = \begin{cases} 1, & F_i \le F^* \\ 1 - \kappa\,\dfrac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35,\ 1]. \qquad (4)$$

4.2. CPA Detection and Give-Way

Each ACTIVE pair is screened with linear CPA. With relative position $\Delta\mathbf{p}$ and relative velocity $\Delta\mathbf{v}$,

$$t^* = -\frac{\Delta\mathbf{p}\cdot\Delta\mathbf{v}}{\|\Delta\mathbf{v}\|^2}, \qquad d_{\text{CPA}} = \big\|\Delta\mathbf{p} + t^*\,\Delta\mathbf{v}\big\|,$$

where t^* is in ticks. A manoeuvre is considered only if $\|\Delta\mathbf{p}\| \le 20$ nm, $d_{\text{CPA}} \le 5$ nm, $0 < t^* \le 12$ ticks, and the give-way ship is not in cooldown. The lower-id ship gives way (turn to starboard); the other stands on. This is a reproducible surrogate for COLREGs stand-on/give-way assignment [10, 11]. The manoeuvre runs only if the warning is delivered:

$$P_{\text{comms}} = \underbrace{\min\big(q(\mathbf{p}_a),\, q(\mathbf{p}_b)\big)}_{\text{weather / storm}} \cdot \underbrace{\min\big(g(F_a),\, g(F_b)\big)}_{\text{fatigue (Eq. 4)}}, \qquad (5)$$

where $q(\mathbf{p})$ is nominal link reliability, reduced to the storm-comms rate when $\mathbf{p}$ is in the storm zone (0.08 in Storm Corridor). A uniform draw below P_{comms} executes the manoeuvre; otherwise geometry alone decides the encounter. A give-way ship receives one temporary waypoint $L_f = 3$ nm ahead and $L_o = 5$ nm to starboard of heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f\,(\sin\psi,\ \cos\psi) + L_o\,(\cos\psi,\ -\sin\psi). \qquad (6)$$

A short message log records the warning exchange; a 3-tick cooldown prevents thrashing.

4.3. Optional Force-Field Track

As an optional, default-off switch, the discrete waypoint of Eq. 6 can be replaced by a continuous repulsion field in the style of Xiao et al. [21]:

$$\mathbf{F}_{\text{rep}} = \sum_{k:\, d_k < d_o} \frac{k_{\text{rep}}}{d_k^p}\,\hat{\mathbf{n}}_k,$$

with exponent $p = 2$ by default. Source effect distances are micro-scale relative to a 15-minute basin tick, so magnitudes are rescaled while keeping the reported head-on to overtaking ratio. All definitive experiments keep this track off.

5. Collision Consequence (SIMCOL)

A **hard collision** is logged when two ACTIVE ships close within 0.10 nm (≈185 m), once per encounter (rising edge). Contacts within 1 nm of a port are ignored as harbour manoeuvring under vessel traffic services (VTS). Damage uses DTU SIMCOL (ship collision damage / survivability) [2]. Absorbed energy follows Pedersen and Zhang [22, 2] on the normal closing component:

$$E_{\text{abs}} = \tfrac{1}{2}\,\frac{M_a'\,M_b'}{M_a' + M_b'}\,V_n^2, \qquad V_n = \big|(\mathbf{v}_b - \mathbf{v}_a)\cdot\hat{\mathbf{n}}\big|, \qquad (7)$$

with added-mass factors $m_{\text{sway}} = 0.85$ and $m_{\text{surge}} = 0.05$ [2]. Minorsky's correlation [23] links energy to damaged volume. Relative penetration t/B drives a survival factor shaped like International Convention for the Safety of Life at Sea (SOLAS) Chapter II-1 probabilistic damage stability [24]:

$$S_i = \begin{cases} 1, & t/B \le r_0 \\ \left(\dfrac{r_1 - t/B}{r_1 - r_0}\right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \ge r_1, \end{cases} \qquad (8)$$

with $r_0 = 0.10$, $r_1 = 0.30$. This is an accepted surrogate: the engine does not carry per-ship residual-stability curves. Expected overboard count is

$$\mathbb{E}[\text{overboard}] = n_b\,\big(1 - S_i\big), \qquad (9)$$

drawn as independent Bernoulli trials. If $S_i < S^\dagger = 0.50$, the struck ship founders and the whole crew enters the liferaft (EVAC, Section 6). If it stays afloat, only the overboard draw enters the man-overboard chain (Section 6.3). Encounter type is diagnostic, from the folded heading angle θ:

$$\text{type} = \begin{cases} \text{side-to-side (overtaking)}, & \theta < 45^\circ \\ \text{front-to-side (crossing)}, & 45^\circ \le \theta < 135^\circ \\ \text{head-on}, & \theta \ge 135^\circ. \end{cases}$$

Near-parallel hits release little energy in Eq. 7; head-on and crossing hits drive deeper penetration.

6. Search and Rescue

Foundering starts the liferaft chain under International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual practice [4]. A rescue asset leaves the nearest port after a mobilisation delay of 2 ticks by default: helicopter (80 kn) beyond 40 nm offshore, otherwise patrol (25 kn). Persons overboard from afloat ships use a separate chain (Section 6.3). Figure 3 shows the vessel state machine. Grounding only reverts position; it does not force evacuation.

6.1. Drift and Detection (MASSIM)

Detection follows the Koopman random-search model used in the MASSIM maritime-search simulation [5, 6]. With sweep width w, search speed v_s, and area A, the cumulative detection probability (CDP) is

$$\text{CDP} = 1 - \exp\left(-\,\frac{n\,w\,v_s\,\tau}{A}\right) = 1 - e^{-C}. \qquad (10)$$

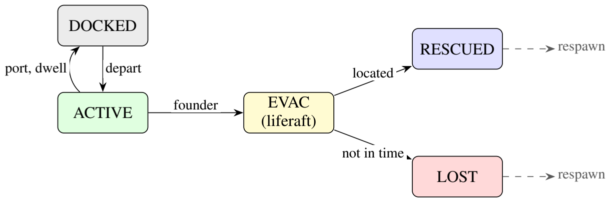

Figure 3: *Vessel state machine. Foundering collision* → EVAC*; SAR yields* RESCUED *(Eq. 10) or* LOST*. Terminal ships respawn to keep fleet size.*

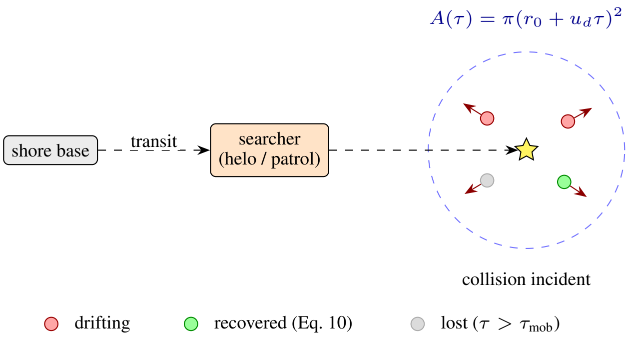

Figure 4: *Man-overboard search. Casualties (Eq. 9) drift separately; one searcher applies Eq. 10 per person.*

Search area grows as $A(\tau) = \pi\,(r_0 + u_d\,\tau)^2$ with initial radius r_0 and drift u_d. Each on-scene tick draws Bernoulli detection at coverage $1 - e^{-\mathrm{d}C}$.

6.2. Seasonal Degradation (Ashrafi)

Asset performance follows Ashrafi et al. [7]. Wind-chill uses

$$\mathrm{wci} = 13.12 + 0.6215\,T - 11.37\,U^{0.16} + 0.3965\,T\,U^{0.16}.$$

Months map to severity groups A–D (May–August mildest; Dec–Feb harshest). Relative transit/search speeds are approximately 1.00/0.95/0.77/0.60 (helicopter) and 1.00/0.80/0.40/0.21 (patrol); boarding times lengthen with severity [7]. Default simulation month is July (group A).

6.3. Man-Overboard Search

Overboard casualties (Eq. 9) form one man-overboard (MOB) incident of persons-in-water, each with its own drift bearing (Fig. 4), following Karatas et al. [5]. One searcher homes on the cluster centroid and applies Eq. 10 independently per person. Anyone not recovered within $\tau_{\text{mob}} = 12$ ticks ($\approx$3 h) is lost. Liferaft endurance is 96 ticks (24 h).

7. External Collision-Frequency Check (IWRAP)

Internal KPIs compare methods. Network collision frequency is checked with the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) Waterway Risk Assessment Program (IWRAP) Mk II [3, 25]:

$$N_c = N_a\,P_c.$$

Head-on, overtaking, and crossing candidate counts follow the usual IWRAP forms (flows Q, beams B, speeds V, Pedersen diameter for crossings [3, 26]). Causation probabilities are the IWRAP defaults: $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$, $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$, $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$. Each batch row stamps N_c from the static synthesised routes, the scenario fleet size, and a method speed scale (weather-aware methods slow traffic). We do not export per-run AIS tracks. For the present network, N_c is about 0.001–0.005 collisions/yr (fleet and speed scale), inside the usual <1/yr/fairway acceptance band.

8. Experimental Design

8.1. Methods and Scenarios

The three methods of Section 2 differ only by fatigue build-rate m (Eq. 3), weather/storm speed awareness (Section 9.2), and preparedness weight α_M below. SIMCOL parameters are shared. Table 2 lists the four scenarios. Each condition uses 30 seeds and 2880 ticks (360 runs).

8.2. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours add 0.25 h per ACTIVE ship per tick. A KPI *fatality* is an unrecovered person-in-water within τ_{mob} (Section 6.3); liferaft LOST is logged separately and does not enter KPI 1. Crew preparedness uses method awareness $\alpha_M \in \{2.0\,(\text{A}),\ 1.3\,(\text{B}),\ 1.0\,(\text{Proposed})\}$:

$$P_{\text{prep}} = \max(0,\ 1 - \alpha_M W)\cdot\max(0,\ 1 - 0.70\,F).$$

Diagnostic counters (collision-type mix, MOB tallies) are logged but not entered into the Holm battery.

8.3. Statistical Analysis

For each (KPI × scenario × method pair) we use a two-tailed Mann–Whitney U test [15] with Cliff's δ [16]. Multiplicity uses Holm's sequentially rejective Bonferroni procedure [27] over the full battery. A result is *significant* if $p_{\text{corr}} < 0.05$ and *confirmed* if also $|\delta| \geq 0.2$. Pairs are (A vs Proposed), (B vs Proposed), and (A vs B). Seeds are matched across conditions.

9. Implementation

The engine is Rust on krABMaga [28]. An Axum HTTP API launches factorial batches (Rayon) and streams progress with Server-Sent Events (SSE). Each finished run writes a JSON Lines (JSONL) tick log (every ten ticks), a manifest, and a SQLite results row. A React workbench (deck.gl / MapLibre) launches batches and replays completed runs (Appendix B).

9.1. Tick Ordering

Within a tick, vessel agents step first (krABMaga order 200), then a post-tick agent (order 1000) runs the shared physics pipeline in Fig. 5; navigation therefore uses weather and avoidance state from the previous tick's post-tick pass.

9.2. Speed Reduction

Baseline B and Proposed interpolate the just-moved position toward the last valid position by

$$f_W = \max\big(0.30,\ 1 - 0.45\,W\big).$$

Baseline A ignores this factor. Inside the storm zone an extra factor applies: 1.0 for Baseline A, 0.45 in Storm Corridor for the other methods (default non-corridor storm factor 0.70 when a storm is enabled).

Table 2: *Scenario summary (30 seeds, 2880 ticks/run).*

Scenario	Key conditions	Ships	Hyp.
Calm Passage	Calm weather; calibration reference (target band ≈0.5–1.5 collisions / 1 k ship-hrs under Baseline A).	20	Calib.
Storm Corridor	Moving storm; storm-comms rate 0.08; non-A speed factor 0.45.	25	H1, H3
Blind Shore	Fleet-wide comms 0.55; tighter avoidance berth ($L_o = 3$ nm).	25	H1
Deep Water Rescue	Sparse fleet; SAR mobilisation 4 ticks; helo/patrol 55/16 kn.	15	H2

Table 3: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	`fatal_per_1k_hrs`	Fatalities ÷ ship-hrs $\times 10^3$	H1, H3
2	`collision_per_1k_hrs`	Collisions ÷ ship-hrs $\times 10^3$	Calib.
3	`evac_activation_rate`	Evac activations ÷ ship-hrs $\times 10^3$	—
4	`survival_ratio`	1− fatalities / crew exposed in collisions	H2
5	`avg_tta_hours`	Mean rescue time-to-arrival	H2
6	`mean_p_prep`	Time-averaged crew preparedness	mech.

10. Results

We report four complementary experiment tiers. (i) The *definitive* factorial uses paper defaults: four scenarios × three methods × 30 seeds × 2880 ticks (360 runs), with IWRAP N_c stamped on every row. (ii) A Tier-1 storm grid varies `storm_comms_success_rate` $\in \{0.04, 0.08, 0.15\}$ and `storm_radius_nm` $\in \{120, 160, 200\}$ on Calm+Storm (five seeds). (iii) The best Tier-1 cell (0.08, 120 nm) is promoted to 30 seeds. (iv) A Tier-2 route-density sweep compares sparse (45), baseline (80), and dense (105) synthesised networks. Confirmation criteria are those of Section 8.3. Figures 6–10 are seaborn summaries of these artifacts.

10.1. Definitive KPI Overview

Table 4 lists mean ± population SD for the three primary KPIs. Figure 6 shows the full per-run distributions. Under Baseline A, Calm Passage collisions sit at 0.60 ± 0.41 per 1 000 ship-hours (inside the 0.5–1.5 calibration band). Storm Corridor raises that rate to 1.48 ± 0.71 (2.48× calm). Blind Shore is the most collision-dense setting (≈ 1.4–1.7), while Deep Water Rescue remains sparse (≈ 0.42–0.49) but carries the only non-negligible fatal base rates.

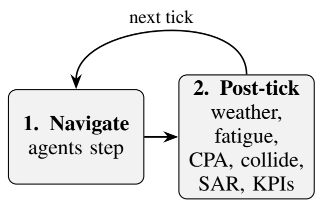

Figure 5: *Macro-tick order. Agents navigate, then the post-tick pipeline advances weather, fatigue, CPA/comms, collisions, SAR/MOB, fleet top-up, and KPIs.*

Table 4: *Definitive batch means ± SD (30 seeds). Collisions and fatals per 1 000 ship-hours; P_{prep} dimensionless.*

Scenario	Method	Collisions	Fatals	P_{prep}
Calm Passage	A	0.60 ± 0.41	0.00 ± 0.00	0.67 ± 0.03
	B	0.51 ± 0.45	0.00 ± 0.00	0.74 ± 0.03
	P	0.48 ± 0.33	0.009 ± 0.035	0.80 ± 0.03
Storm Corridor	A	1.48 ± 0.71	0.00 ± 0.00	0.16 ± 0.03
	B	1.40 ± 0.57	0.002 ± 0.013	0.21 ± 0.03
	P	1.07 ± 0.51	0.002 ± 0.012	0.30 ± 0.04
Blind Shore	A	1.66 ± 0.70	0.005 ± 0.017	0.47 ± 0.05
	B	1.57 ± 0.70	0.002 ± 0.012	0.55 ± 0.03
	P	1.41 ± 0.61	0.007 ± 0.020	0.64 ± 0.03
Deep Water	A	0.49 ± 0.43	0.038 ± 0.094	0.48 ± 0.05
	B	0.44 ± 0.35	0.070 ± 0.150	0.55 ± 0.04
	P	0.42 ± 0.41	0.023 ± 0.056	0.65 ± 0.04

10.2. Hypotheses H1–H3

Table 5 breaks the pre-registered contrasts down by scenario. **H1** and **H2** are not confirmed anywhere after Holm: fatal rates and survival ratios sit near floor/ceiling in Calm, Storm, and Blind Shore ($|\delta| \le 0.07$). Deep Water Rescue has the only material fatal mass; Proposed reduces mean fatals from 0.038 (A) to 0.023 (≈39 %), but $\delta \approx 0.01$ and $p_{\mathrm{corr}} = 1$. **H3** is likewise null (Storm fatals 0.0024 vs 0.0022). Figure 7 places these effect sizes next to the collision and preparedness contrasts: the only consistently confirmed A→P lever is P_{prep}.

10.3. Mechanism: Preparedness and Storm Collisions

Mean P_{prep} rises monotonically A<B<P in every scenario, with all twelve pairwise contrasts confirmed ($|\delta| \in [0.72, 1.00]$; Fig. 8, left). That matches the design of α_M and $m(\text{method})$. On Storm Corridor collisions (Fig. 8, right), Proposed cuts the Baseline A mean from 1.48 to 1.07 (≈28 %, $\delta = 0.32$, raw $p \approx 0.035$), but the contrast does not survive family-wise Holm correction. Blind Shore shows a similar directional drop (1.66 → 1.41) without confirmation.

10.4. IWRAP Network Frequency

Every definitive run carries an IWRAP Mk II stamp N_c from the static route network, fleet size, and method speed scale (Table 6). Values lie in 0.001–0.005 collisions/yr—well inside the <1/yr/fairway band—and decrease for weather-aware methods because slower traffic reduces encounter opportunity. Stamps are identical for Storm and Blind at equal fleet ($N = 25$), as expected for a geometry-only check.

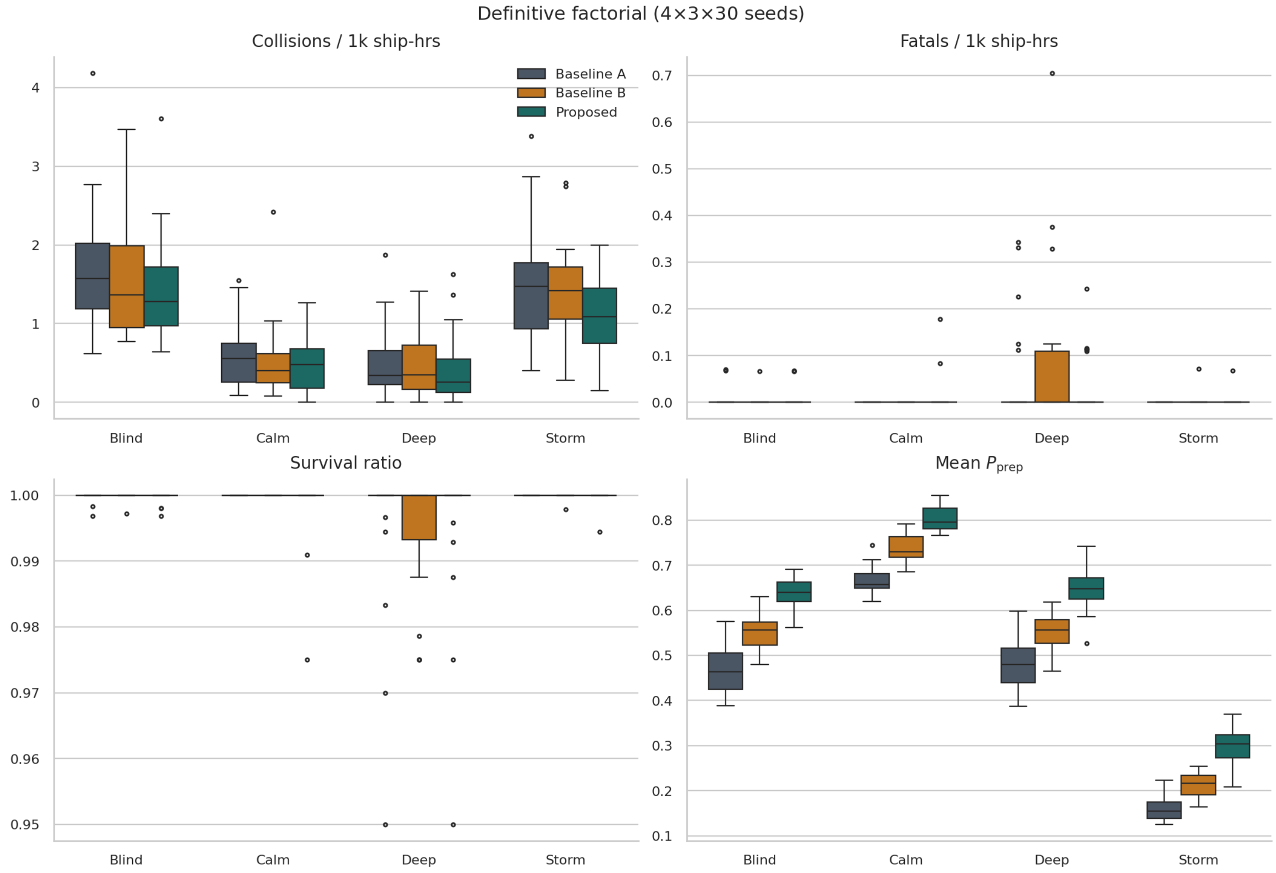

Figure 6: *Definitive factorial KPI distributions (30 seeds). A/B/P denote Baseline A, Baseline B, and Proposed.*

Table 5: *Pre-registered hypotheses by scenario (Proposed vs A for H1/H2; vs B for H3).*

Hyp.	Scenario	Means (ref→P)	δ	Verdict
H1	Calm	$0.000 \to 0.009$	-0.07	Not conf.
H1	Storm	$0.000 \to 0.002$	-0.03	Not conf.
H1	Blind	$0.005 \to 0.007$	-0.03	Not conf.
H1	Deep	$0.038 \to 0.023$	$+0.01$	Not conf.
H2	all four	survival ≥ 0.995	$\lvert\delta\rvert \leq 0.07$	Not conf.
H3	Storm	$0.0024 \to 0.0022$	$+0.00$	Not conf.

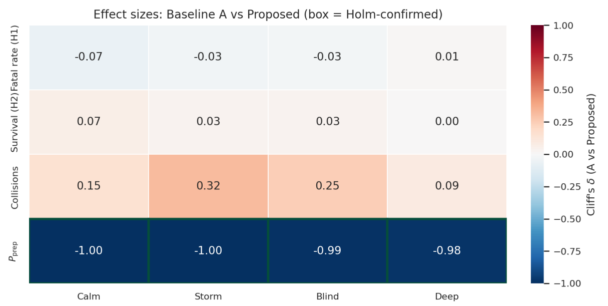

Figure 7: *Cliff's δ for Baseline A vs Proposed. Green boxes mark Holm-confirmed cells ($|\delta| \geq 0.2$, $p_{\text{corr}} < 0.05$).*

10.5. Storm Hyperparameter Sensitivity

Figure 9 summarises the Tier-1 grid (five seeds; Calm collisions are invariant to storm knobs, as expected). Storm/Calm ratios span roughly 2.0–3.5×. The paper default (0.08, 160 nm) yields 2.39×. The calibration objective prefers (0.08, 120 nm) at 2.64× with Calm A collisions ≈ 0.66. Promoting that cell to 30 seeds keeps a clear weather contrast (2.27×) but does *not* unlock H1–H3: fatal reductions remain null or reverse, confirming that fatal sparsity—not storm geometry alone—limits hypothesis power.

10.6. Route-Density Sensitivity

Table 7 and Figure 10 compare sparse (45), baseline (80), and dense (105) networks under paper storm defaults (five seeds). Sparse traffic sits nearest the dual calibration targets (Calm A ≈ 0.91, Storm/Calm ≈ 2.66×). Denser lanes lower absolute collision rates and compress the weather contrast (1.85×), so the storm stressor becomes harder to see. Method ordering on P_{prep} is unchanged. We therefore retain the committed 80-route baseline for the definitive claim, and treat sparse/dense as a structural sensitivity rather than a retune.

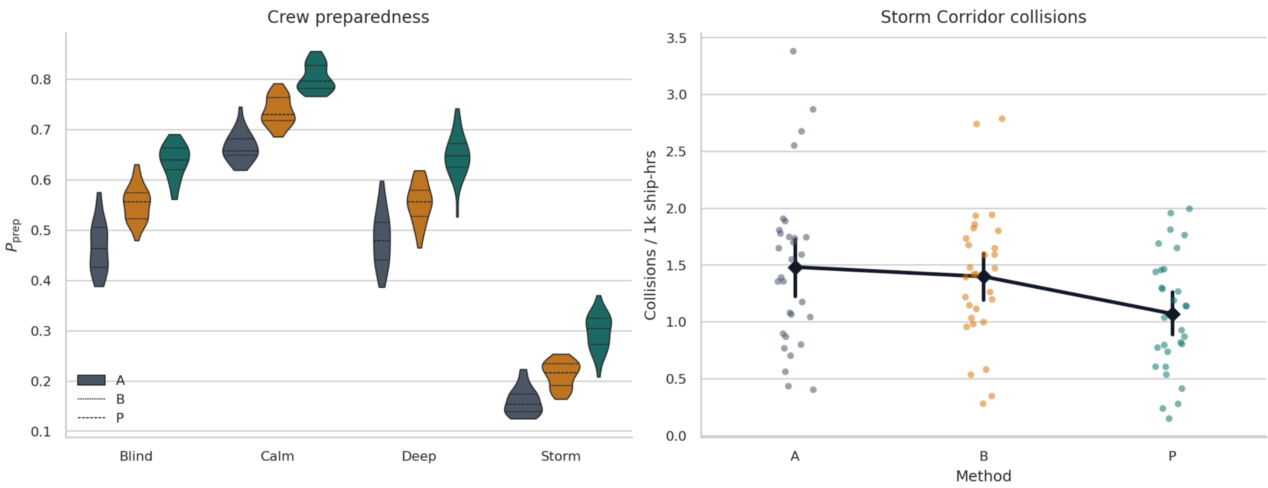

Figure 8: *Left:* P_{prep} *violins (definitive). Right: Storm Corridor collision strip with mean* $\pm$*95 % CI.*

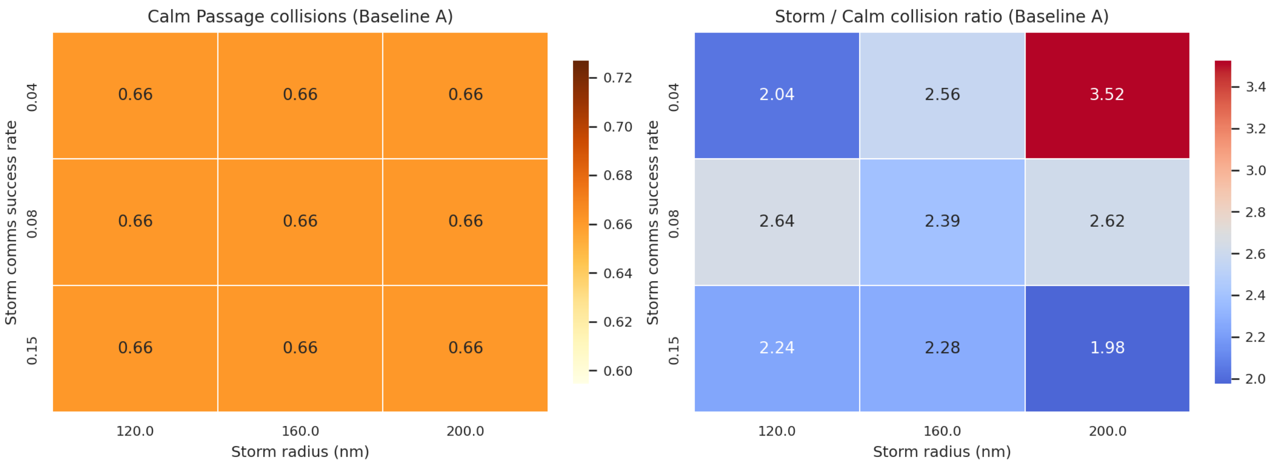

Figure 9: *Tier-1 storm grid (Baseline A, five seeds). Left: Calm collisions. Right: Storm/Calm ratio (diverging around the* 2.7$\times$ *target).*

Table 6: *Mean IWRAP* N_c *(collisions/yr) stamped on the definitive batch.*

Scenario (N)	A	B	P
Calm (20)	0.0033	0.0028	0.0025
Storm / Blind (25)	0.0051	0.0043	0.0040
Deep (15)	0.0018	0.0016	0.0014

Table 7: *Tier-2 route density (five seeds): Baseline A collisions and Storm/Calm ratio.*

Network	Routes	Calm A	Storm/Calm
Sparse	45	0.91	2.66$\times$
Baseline	80	0.66	2.39$\times$
Dense	105	0.48	1.85$\times$

11. Discussion

The core claim is Eq. 5: a warning succeeds only when weather and both crews allow it. Across the definitive factorial, the storm grid, the 30-seed promote, and the route-density sweep, the batch confirms the preparedness lever and a directional storm-collision benefit, but not the pre-registered fatal/survival thresholds. Fatal events remain rare outside Deep Water Rescue, so H1–H3 are under-powered even when point estimates move in the expected direction. Storm radius/comms and network density shift absolute rates and weather contrast, yet they do not create a fatal-signal where the MOB window rarely elapses. IWRAP stamps (0.001–0.005/yr) place the synthesised network in a credible low-frequency band. The survival factor S_i (Eq. 8) is a SOLAS-shaped surrogate, not a full residual-stability calculation. Main limitations: synthesised routes trade some traffic realism for control; the lower-id give-way rule simplifies COLREGs; force-field avoidance is off by default; IWRAP uses static routes rather than per-run tracks. Calm Passage calibration under Baseline A sits near published AIS near-miss bands [12, 13]. Longer runs or collision-rate primary endpoints would raise power on fatal contrasts.

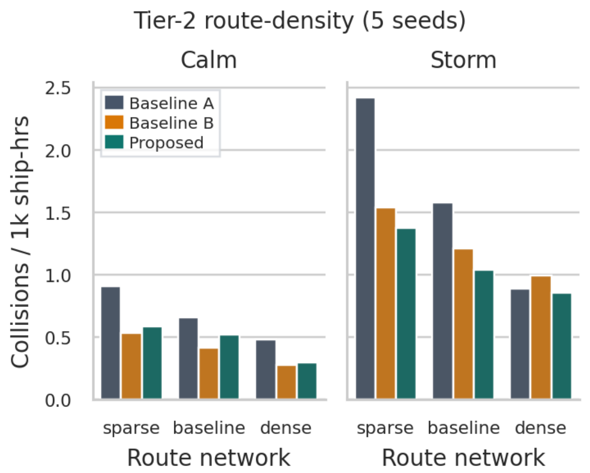

Figure 10: *Tier-2 collisions by route network, method, and scenario (five seeds).*

12. Conclusions

Section 2 asked whether linking fatigue management to warning reliability reduces fatal outcomes relative to classical, weather-unaware practice. On the pre-registered fatal and survival endpoints, the answer is *no*—not at the stated thresholds, under the definitive 360-run design.

H1 (Proposed cuts fatal rate by $\geq 20\,\%$ vs Baseline A) is **not confirmed**. In Calm, Storm, and Blind Shore the fatal KPI is essentially at the floor, so relative reductions are undefined or negligible ($|\delta| \leq 0.07$, $p_{\text{corr}} = 1$). Deep Water Rescue is the only setting with a material base rate; Proposed lowers the mean from 0.038 to 0.023 ($\approx 39\,\%$), but the contrast fails Holm and Cliff's criteria ($\delta \approx 0.01$).

H2 (Proposed improves survival by $\geq 15\,\%$ vs Baseline A) is **not confirmed**. Survival ratios remain ≥ 0.995 in every scenario; ceiling effects leave no room for a $15\,\%$ gain ($|\delta| \leq 0.07$).

H3 (under Storm Corridor, Proposed beats Baseline B on fatals) is **not confirmed**. Means are 0.0022 vs 0.0024 ($\delta \approx 0$, $p_{\text{corr}} = 1$).

What *is* supported, and answers the broader modelling question in Section 2, is the *mechanism*: coupling weather and fatigue into P_{comms} (Eq. 5) reliably raises crew preparedness (P_{prep} confirmed in all twelve method pairs, $|\delta| \geq 0.72$) and directionally reduces Storm Corridor collisions versus Baseline A ($\approx 28\,\%$, $\delta = 0.32$, not Holm-significant). Storm and route-density sensitivities change absolute rates and weather contrast but do not overturn the H1–H3 verdicts. Future work should either lengthen runs / raise traffic until MOB fatalities are powered, or re-register collision rate and P_{prep} as primary endpoints.

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A. Default Constants

Table 8 lists principal defaults (overridable in configuration). Scenario overrides that differ from these defaults appear in Table 2.

Weather-field process (Mixed preset): regime flips $p_{cs} = 0.03$, $p_{sc} = 0.08$; Calm/Storm reversion targets $(0.30, 0.80, 0.25)$ / $(0.72, 0.30, 0.75)$; up to 8 storm cells; AR(1) coefficients $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$, $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$; gradient cap 0.12. Calm and Stormy presets scale the flip rates.

B. Playback Workbench

The optional React 19 / Vite workbench is a batch-first client for the Axum API: it does not run the ABM in the browser. Figure 11 shows the shell—a permanent left panel beside a full-bleed deck.gl map over a MapLibre basemap. Under RUNS, the operator selects scenarios and methods, seed/tick counts, and an optional JSON config override, then posts a factorial batch; progress arrives over SSE while finished seeds appear in a completed-run list. Selecting a seed loads its JSONL tick log and enables the pinned playback dock (play/pause, speed, scrubber). Map layers render vessels and ports, AIS route context, weather/storm overlays, collisions and wrecks, and SAR/MOB entities, with camera and layer toggles in the map chrome.

Figure 12 shows the COMMS tab: bridge messages up to the scrubber (collision warnings with CPA/TTA, give-way, resume-route, keep-distance). STATISTICS and TELEMETRY (not shown) pull pairwise Mann–Whitney / Holm summaries and KPI time series for the active batch. The GUI is for inspection and debugging—all hypothesis numbers in the main text come from exported batch JSON / SQLite, not from the browser.

Table 8: *Selected engine default constants.*

Constant	Default	Role
$\Delta t, T$	0.25 h, 2880	tick length; ticks per run (30 d)
N vessels	20	Calm Passage; Storm/Blind 25, Deep Water 15
n crew	10	per vessel
weather grid; (w_s, w_v, w_u)	50×50; $(1.0, 0.5, 0.8)$	hazard field and fusion weights
q (nominal link)	1.0	fleet-wide comms rate (Blind Shore 0.55)
warn / CPA / TTA gate	20 / 5 nm / 12 tk	avoidance gate
L_f, L_o; cooldown	3 / 5 nm; 3 tk	give-way offsets; re-manoeuvre lockout
waypoint reach	0.5 nm	route / avoidance waypoint capture
r_{coll}; port exclusion	0.05 nm; 1 nm	hit at $2r$=0.10 nm; harbour water
follow vicinity / gap; same-dest.	3 / 0.5 nm; 5 nm	speed-match; endpoint match
dwell	8–48 tk	port stay
m(A/B/Proposed)	$1.00/0.82/0.65$	fatigue build-rate (Eq. 3)
$(\beta_{\text{base}}, \beta_W, \beta_c)$	$0.0015, 0.0040, 0.0008$	fatigue rates
recovery; F^*, κ	0.025/tk; 0.40, 0.65	dock recovery; comms knee
α_M; fatigue weight	$2.0/1.3/1.0$; 0.70	P_{prep} awareness / F term
f_W (B, Proposed)	$\max(0.30,\ 1 - 0.45\,W)$	weather speed damping; A ignores
storm radius / drift	120 nm / 0.30 nm/tk	generic storm; Corridor radius 160 nm
storm-comms	0.30 (0.08 Corridor)	link rate inside storm zone
storm speed factor	0.70	A: 1.0; Corridor non-A: 0.45
force-field track	off	optional Xiao-style repulsion
$(m_{\text{sway}}, m_{\text{surge}})$	$0.85, 0.05$	SIMCOL added mass
Minorsky a, b	$47.2, 32.7$	energy–volume (MJ, m^3)
$\ell/L_{pp}, \tau$	0.12, 0.08 m	damage length / side steel
$r_0, r_1, S^\dagger$	$0.10, 0.30, 0.50$	S_i knee / loss / founder
helo / patrol; cutoff	80 / 25 kn; 40 nm	SAR speeds; asset-selection range
Δt_{mob}; arrival r	2 tk; 1 nm	mobilisation; datum arrival
w, r_0^{SAR}, u_d	5, 2 nm, 0.2 nm/tk	MASSIM sweep / datum / drift
survival / τ_{mob}	96 / 12 tk	liferaft (24 h); MOB (3 h)
MOB scatter; searchers	0.5 nm; 1	person-in-water spread; search team
sim. month	July (Ashrafi A)	seasonal SAR degradation group
P_c (IWRAP)	0.5–1.3×10^{-4}	head-on / overtaking / crossing
route weights (w_h, w_d, w_ℓ)	$(3, 4, 2.5)$	offline A^* cost (Section 3.3)
UKC buffer m_s	2 m	navigable-depth margin

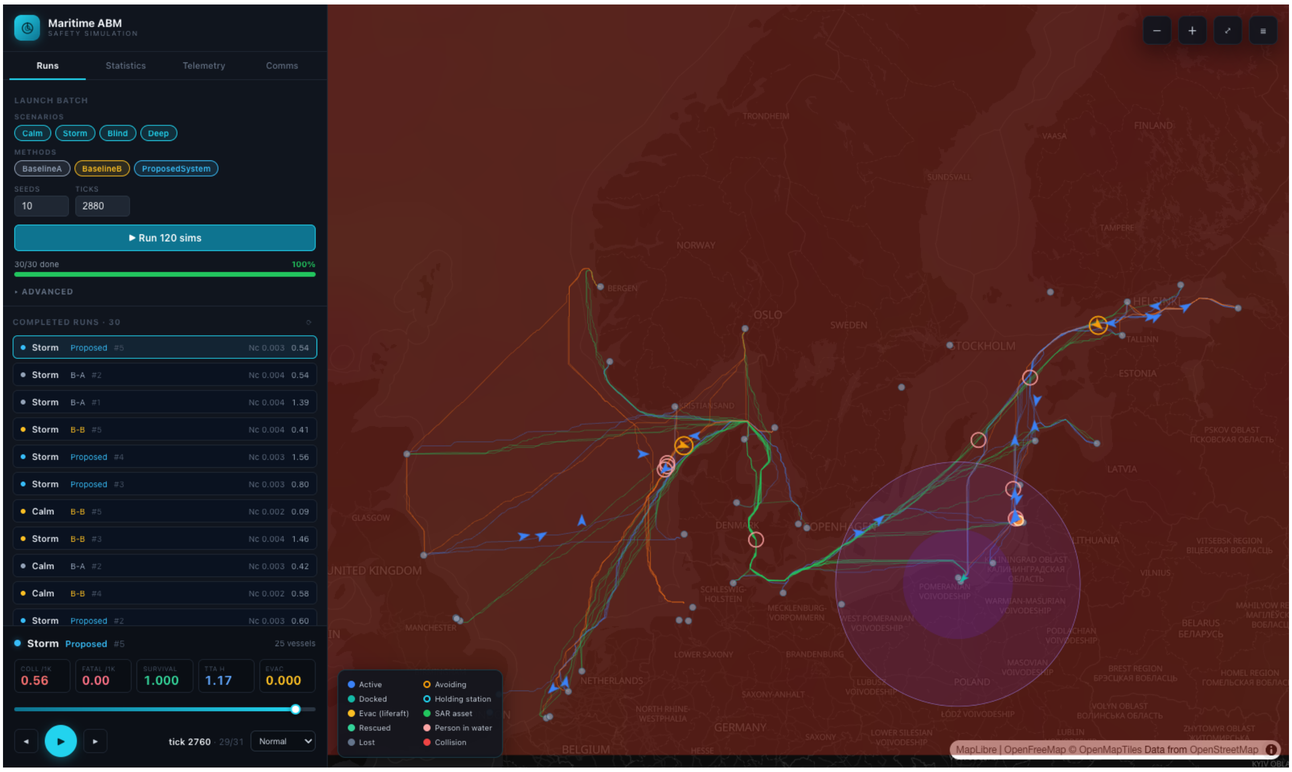

Figure 11: *Workbench shell scrubbed mid-run (Storm Corridor, Proposed, seed 5, tick 2760): live KPI tiles, vessel tracks / avoidance rings / collisions on the map, and the storm disc over the southern Baltic.*

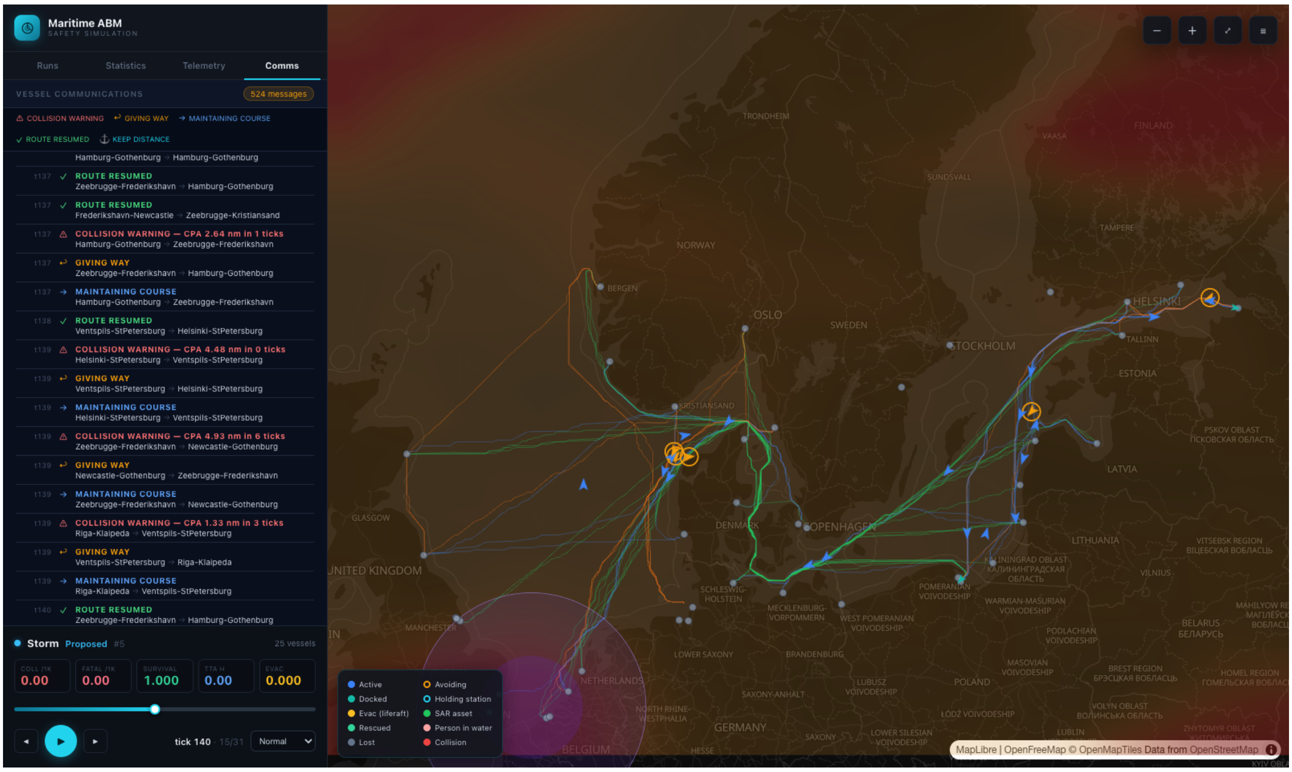

Figure 12: *Comms log for the same Storm Corridor run (hundreds of VHF-style messages with CPA warnings and give-way acknowledgements), with the playback map still showing routes and the moving storm.*